

Humanoid World Model

Students: Hung Nguyen, Khang Le, Minh Tran

Introduction

Motivation - The Need for World Models

Do We Have Enough Data For Humanoid Robot?



Do We Have Enough Data For Humanoid Robot?

Usually we use real world data,
language, vision, audio ...

But is that enough for robots?

Do We Have Enough Data For Humanoid Robot?

Datasets

 **1M Episodes** from **311 Scenes** 
 **34 Research Labs** across **21 Institutions** 

22 Embodiments



527 Skills



60 Datasets

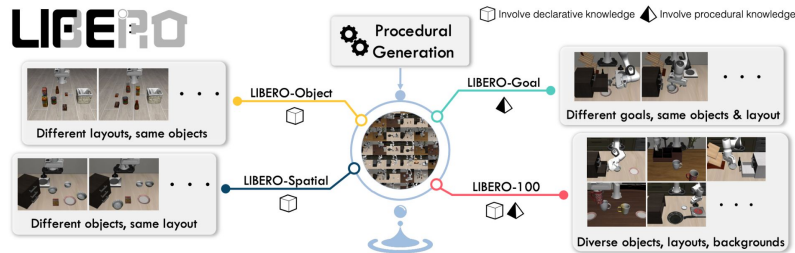
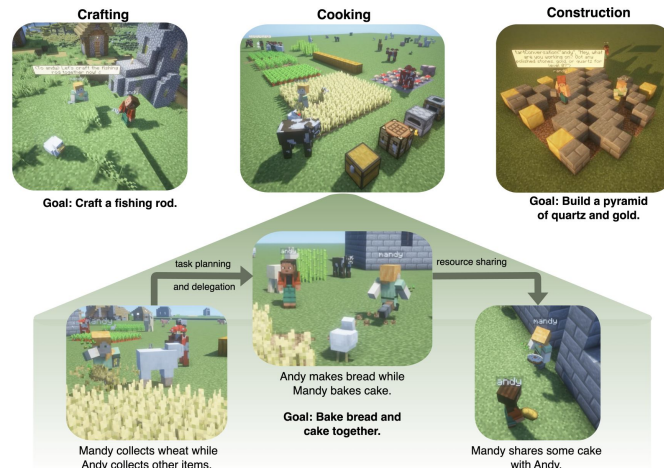


1,798 Attributes • 5,228 Objects • 23,486 Spatial Relations

Real-world Data

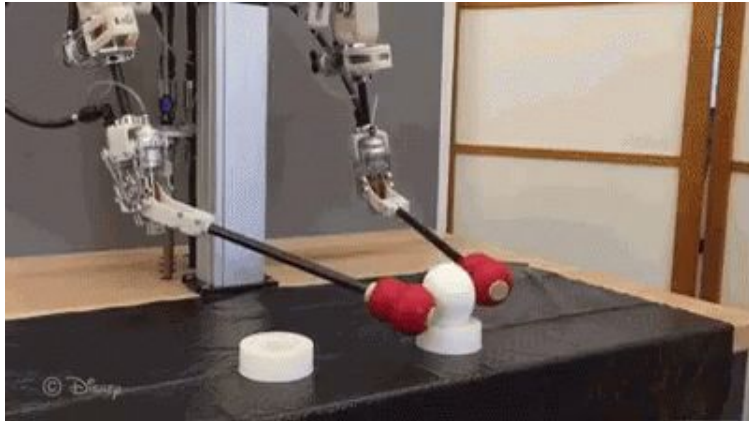
DROID
Distributed Robot
Interaction Dataset

 **76k Episodes**
 **564 Scenes**
 **52 Buildings**
 **13 Institutions**
 **86 Tasks / Verbs**

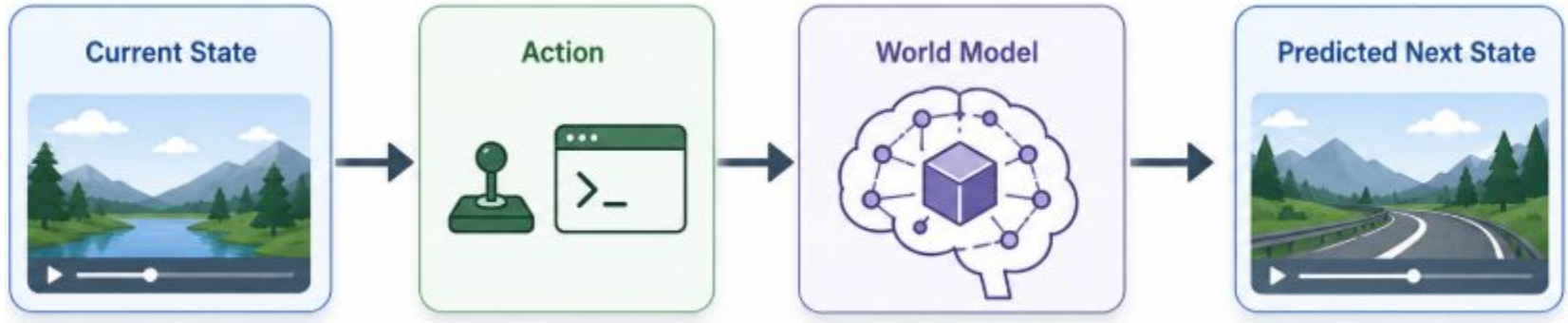


Simulated Data

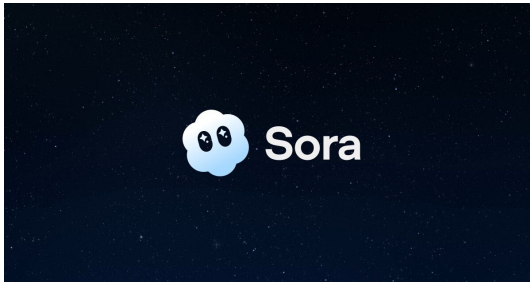
Safety Problems



World Model - A model that understand the world



Applications

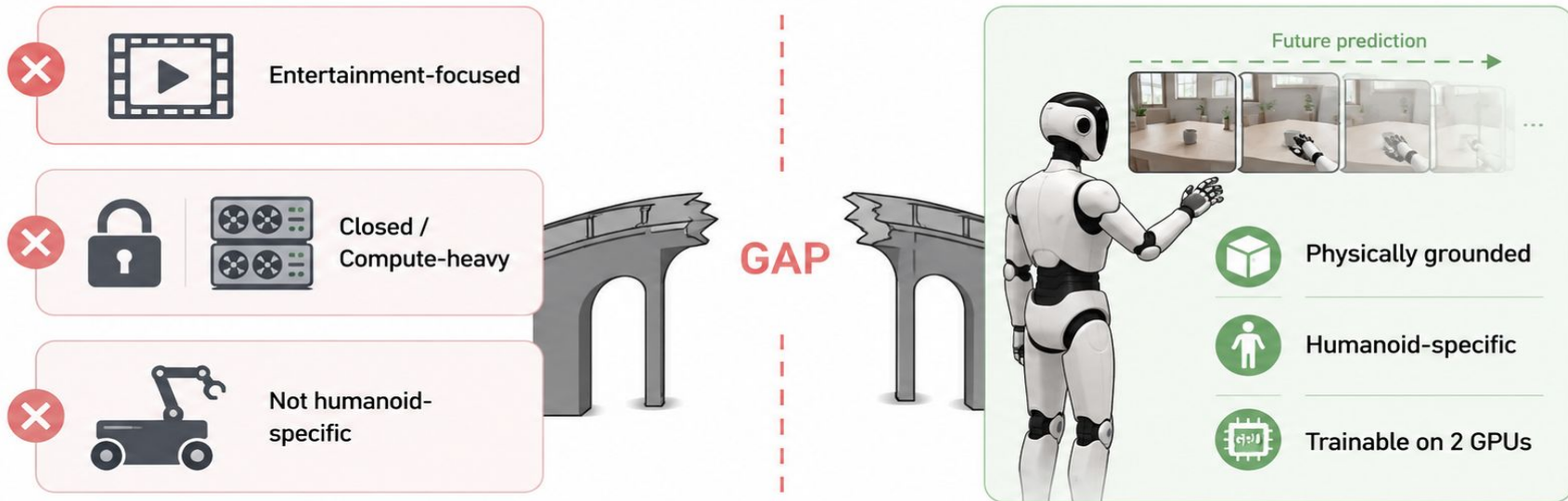


Video Generation



Games (Genie 3)

Research GAP

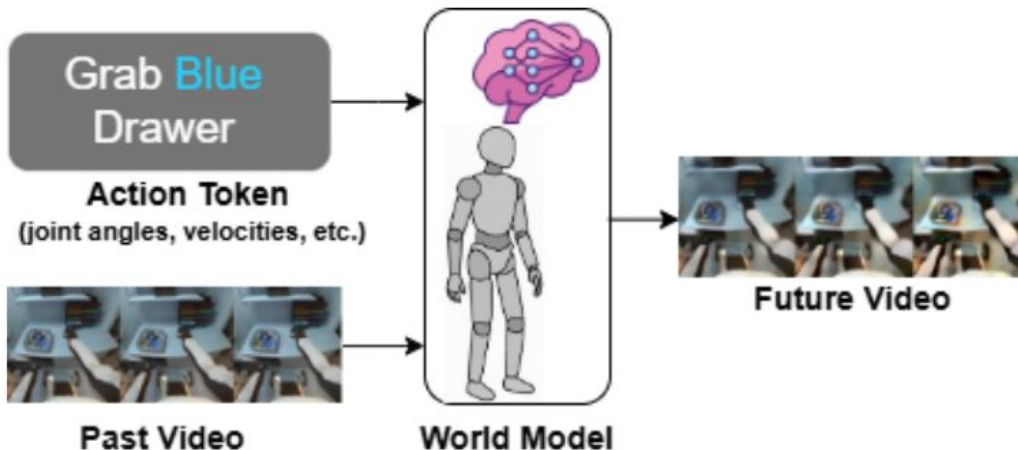


Need lightweight open world models for humanoid robotics.

Introduction to HWM



Humanoid World Models (HWM) that forecast future egocentric video conditioned on humanoid control tokens

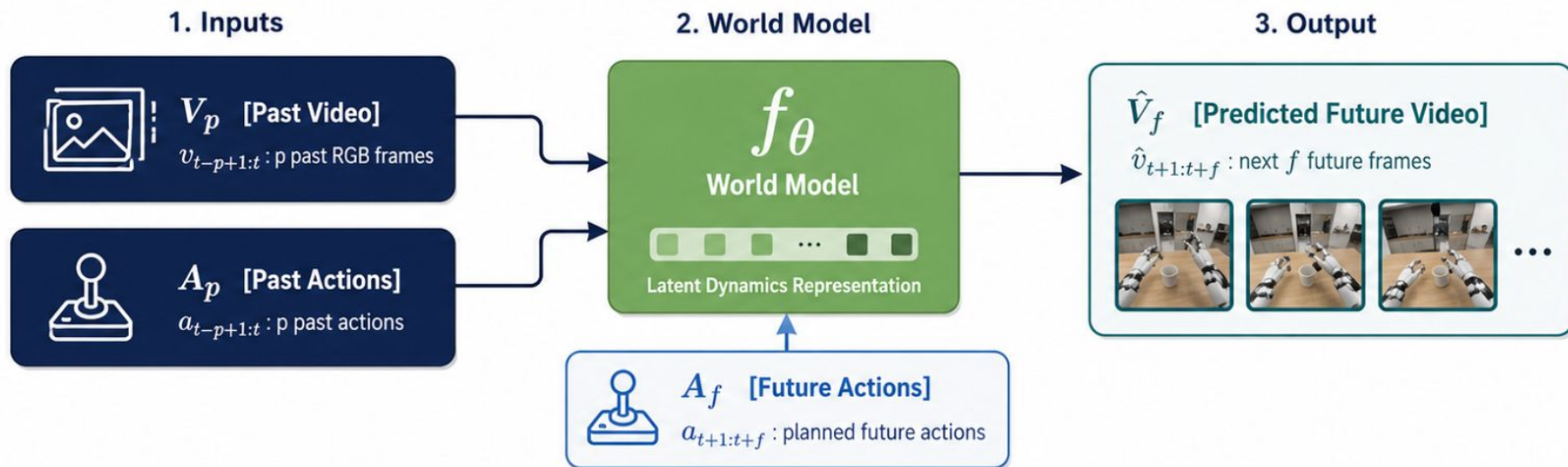


Problem Formulation

At each time step t , condition on past observations and planned actions to predict future egocentric frames.

$$\text{Learn } f_{\theta} : (V_p, A_p, A_f) \rightarrow \hat{V}_f$$

$$V_p = v_{t-p+1:t} \quad A_p = a_{t-p+1:t} \quad A_f = a_{t+1:t+f}$$



**How do we generate
future video?**

Masked Video Modelling

Future Prediction as Token Filling

Inputs

Past video V_p
Past actions A_p
Future actions A_f

VQ-VAE Tokenizer

Compress video into
discrete latent tokens
 $V_p, V_f \rightarrow L_p, L_f$

Mask Future Latents

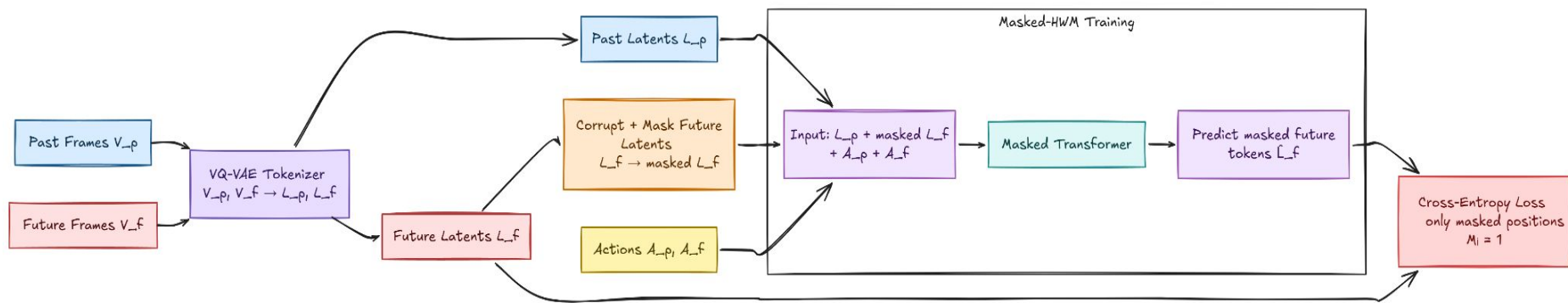
Mask future
tokens in L_f

Masked Transformer

Predict masked future
tokens using L_p, A_p and A_f

Decode

Recover future video
 $\hat{L}_f \rightarrow \hat{V}_f$



Training Objective and Inference

Original L_f

12

48

7

91

5

23

Masked input

12

[M]

7

[M]

5

[M]

Model output

12

48

7

91

5

23

Only the masked positions contribute to the loss.

Cross-Entropy Loss:

$$\mathcal{L} = -\mathbb{E}_{\mathbf{M}} \left[\sum_i \mathbf{M}_i \log p(\hat{L}_i | \mathbf{L}_f) \right]$$

$\mathbf{M}_i = 1$ only for masked positions in future latents L_f .

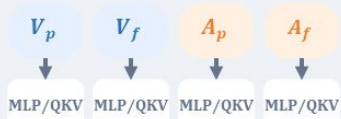
Inference:

Mask all future tokens L_f , predict tokens in parallel, then refine low-confidence tokens over K steps.

Efficiency via Parameter Sharing

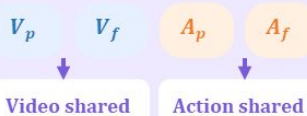
Base Block

Separate parameters for each stream



Modality Sharing

Share within similar modalities



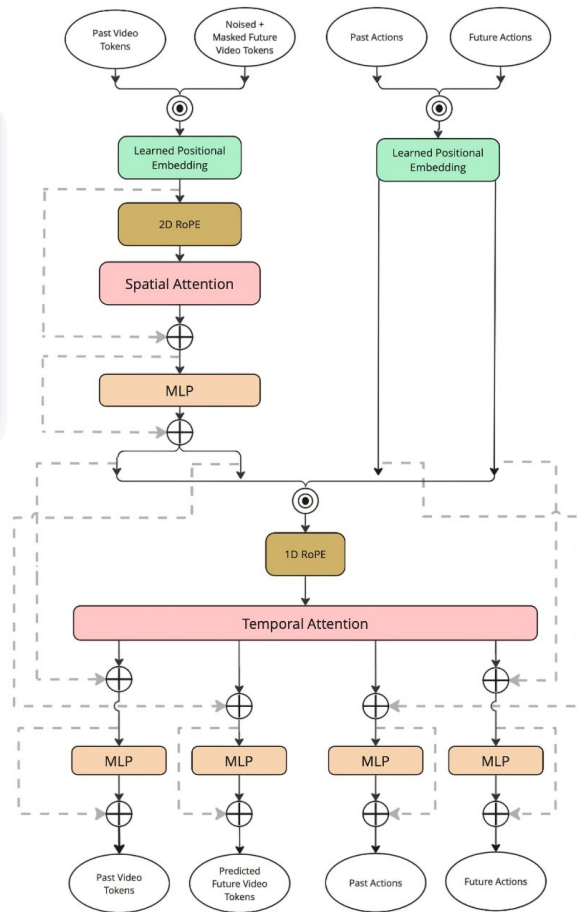
Full Sharing

Share major components across all streams



Masked-HWM parameter count

Base		0.321B	
Modality		0.237B	-26%
Full		0.195B	-39%

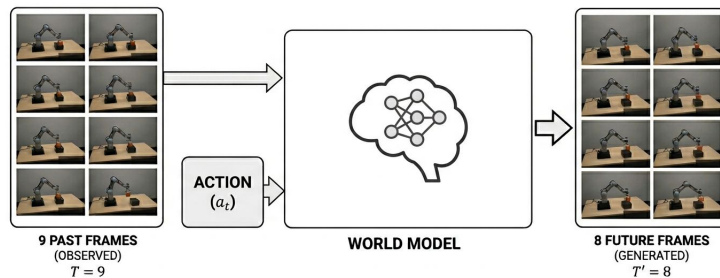
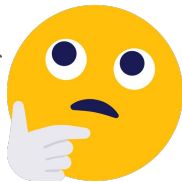


Base Block

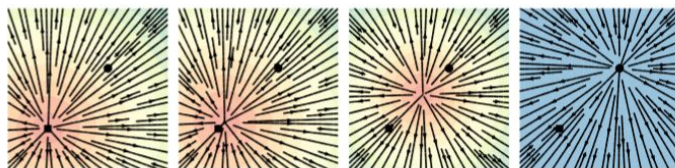
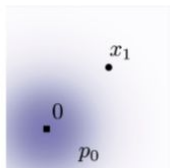
Flow Matching

Why do we need Flow Matching?

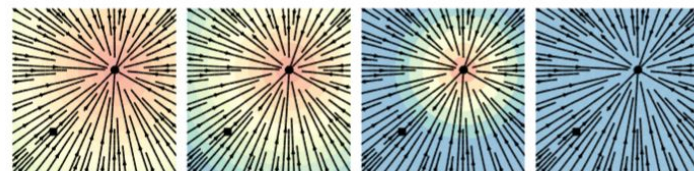
Diffusion too slow!!
Alternative??



Link: <https://gemini.google.com/app/9128bdc96fe3cb13>



Diffusion path – conditional score function



OT path – conditional vector field

<https://arxiv.org/pdf/2210.02747>

Core Idea: Velocity Field

Line from noise to real data

$$X_t = t \cdot X_1 + (1 - (1 - \sigma_{min})t) \cdot X_0, \quad t \in [0, 1]$$

Velocity

$$V_t = \frac{dX_t}{dt} = X_1 - (1 - \sigma_{min}) \cdot X_0 \approx X_1 - X_0$$

Model learn to predict

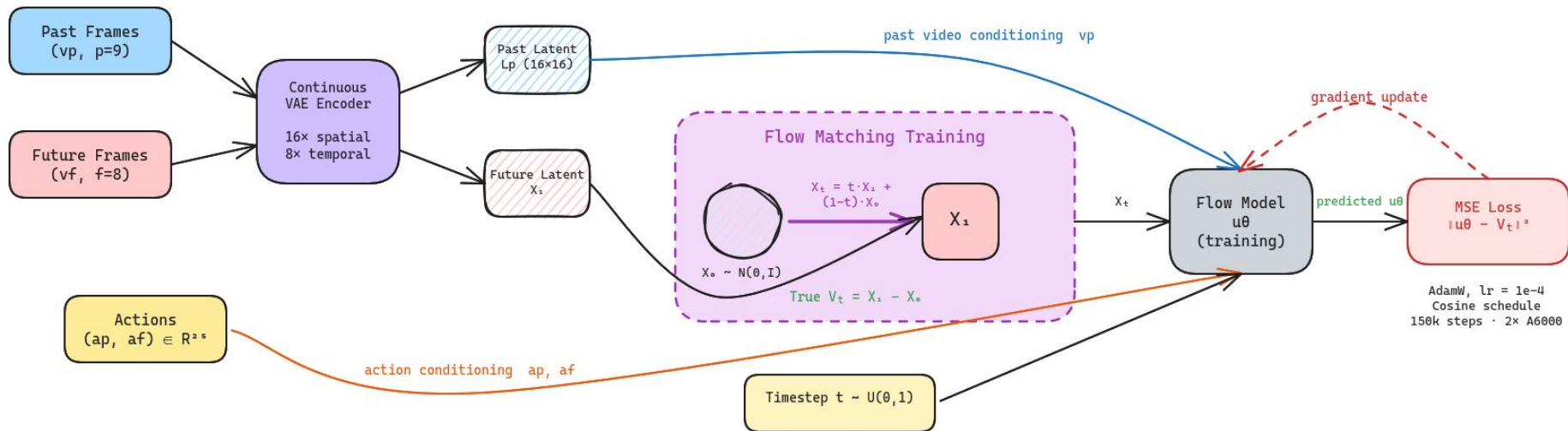
$$u_\theta(X_t, v_p, a_p, a_f, t) \approx V_t$$

With the training loss

$$\mathcal{L} = \mathbb{E} \left[\|u_\theta(X_t, v_p, a_p, a_f, t) - V_t\|^2 \right]$$

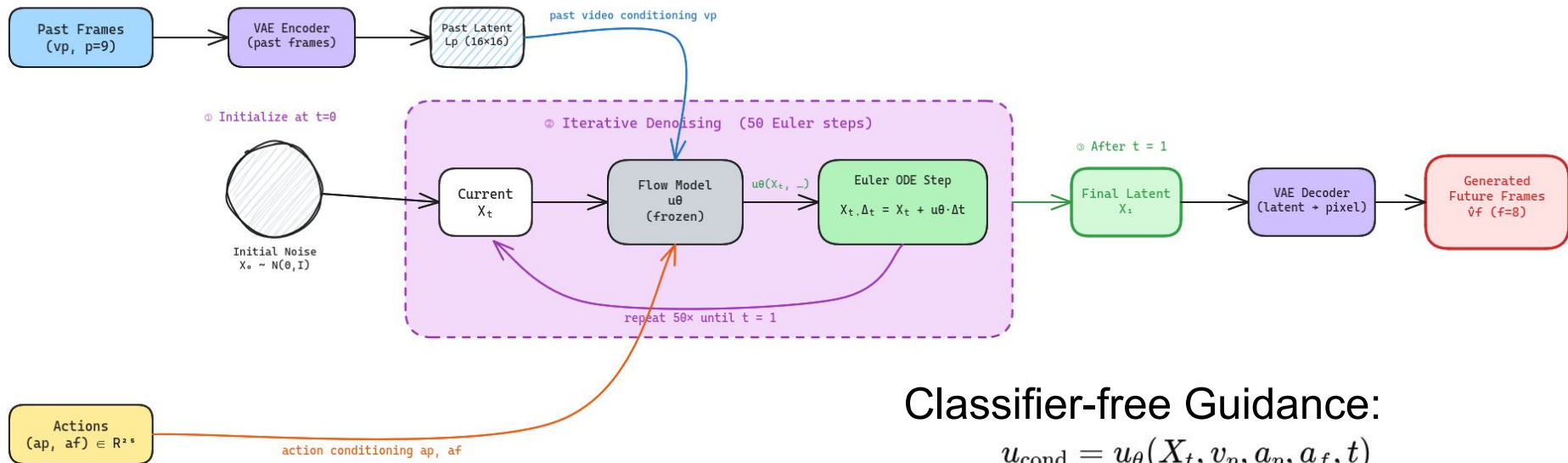
Training & Inference

Flow-HWM Training Pipeline



Training & Inference

Flow-HWM Inference Pipeline



Classifier-free Guidance:

$$u_{\text{cond}} = u_\theta(X_t, v_p, a_p, a_f, t)$$

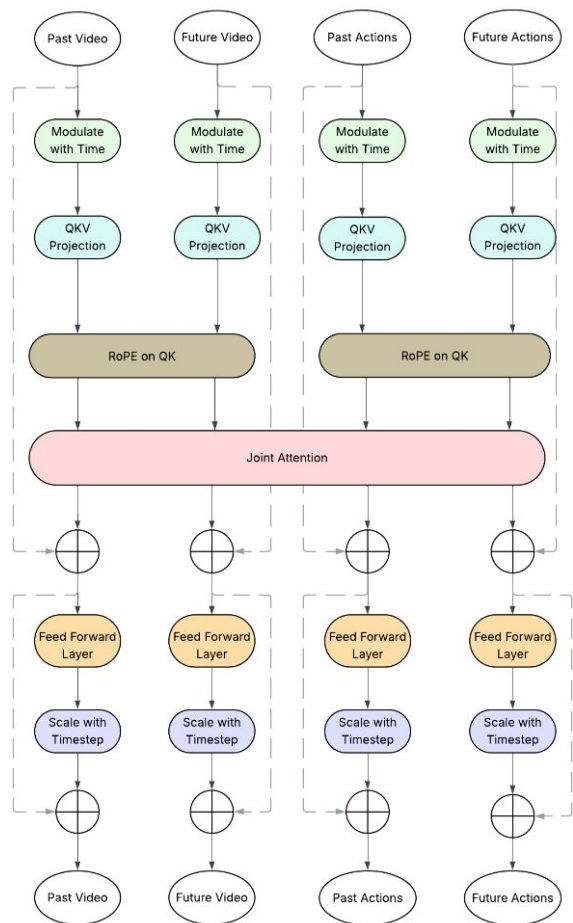
$$u_{\text{uncond}} = u_\theta(X_t, \emptyset, \emptyset, \emptyset, t)$$

$$u_{\text{guided}} = u_{\text{uncond}} + 3.0 \cdot (u_{\text{cond}} - u_{\text{uncond}})$$

$$X_{t+\Delta t} = X_t + u_{\text{guided}} \cdot \Delta t$$

Base Block of Flow-HWM

- Each token stream (past/future video and actions) uses separate weights for timestep modulation, QKV projection, and feedforward MLPs.
- Joint Attention integrates all streams.
- RoPE is applied by modality: 3D for video tokens, 1D for action tokens.



Impacts

Four core impacts:

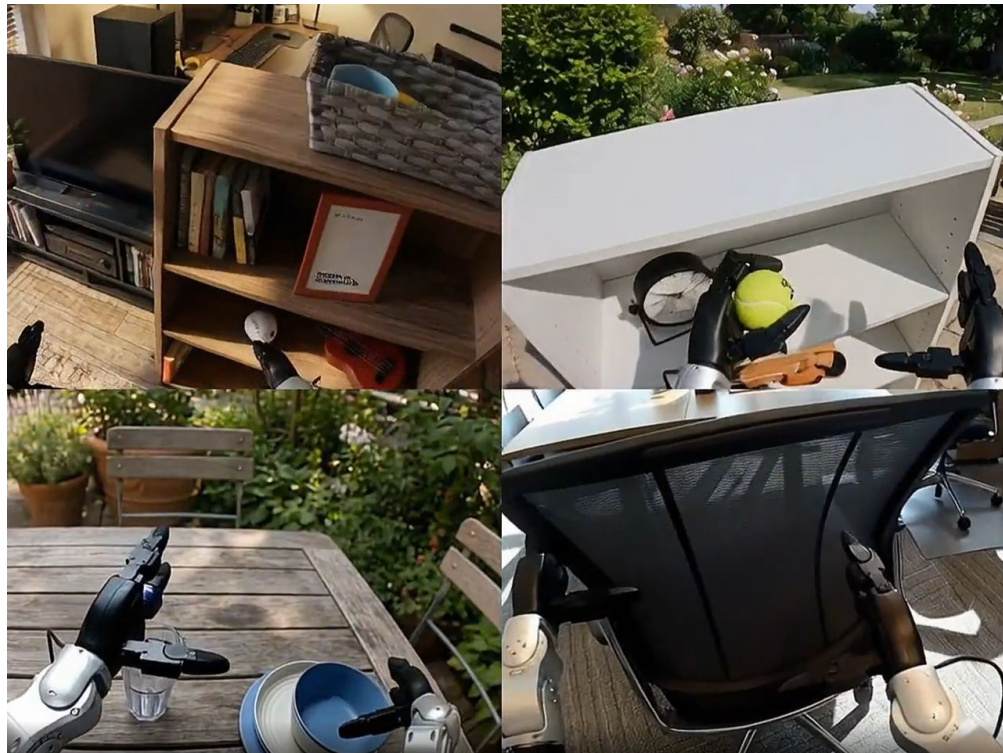
1. Democratizing AI research via resource optimization
2. Bridging the Sim-to-Real gap with synthetic data
3. Pioneering action-conditioned video generation in robotics
4. World Model as An Environment for Policy Evaluation

Democratizing Research & Optimizing Resources

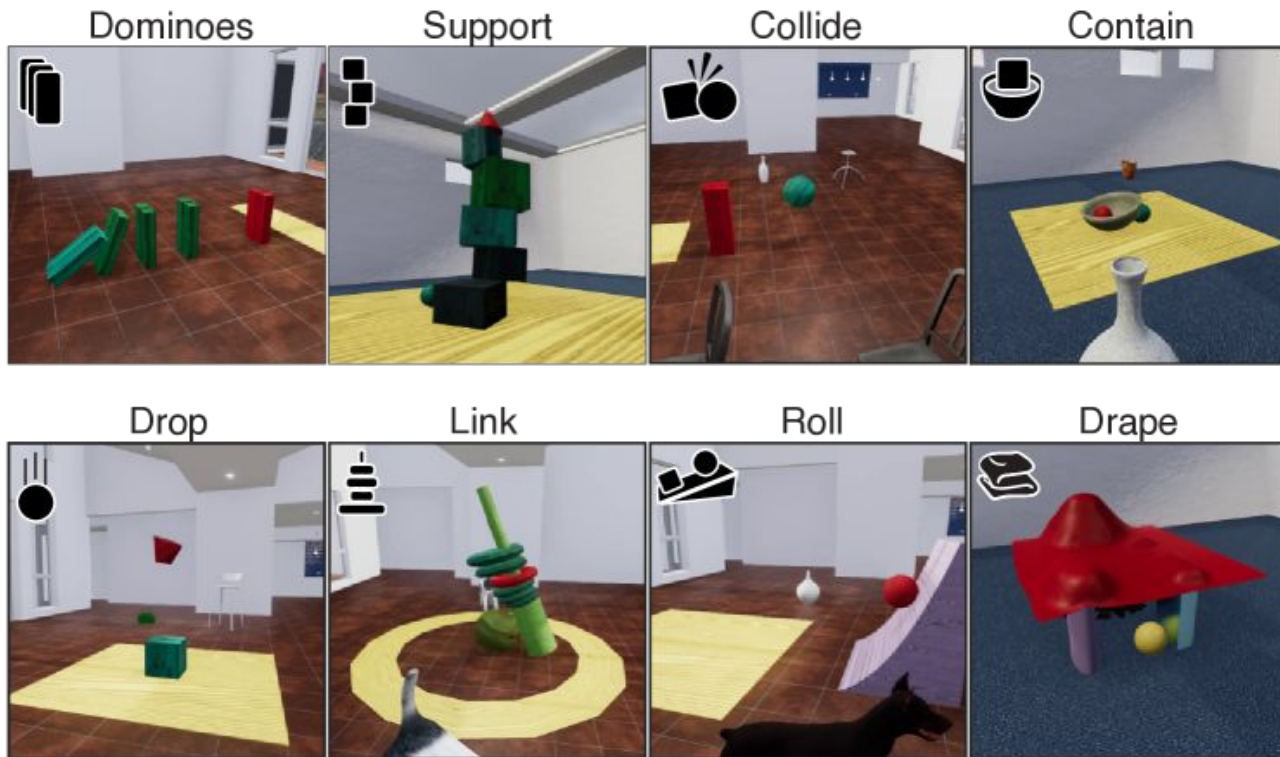
METRIC	SPLIT ATTENTION	BASE BLOCK	MODALITY SHARING	FULL SHARING
MODEL SIZE (BILLION)	0.220	0.321	0.237	0.195
PEAK GPU MEMORY (GB)	2.22	2.63	2.30	2.12
SAMPLES PER SECOND	2.09	2.27	2.25	2.36
FID	15.31	10.13	11.67	14.21
PSNR (DB)	29.37	29.02	28.97	28.66

METRIC	SPLIT ATTENTION	BASE BLOCK	MODALITY SHARING	FULL SHARING
MODEL SIZE (BILLION)	0.944	1.36	0.886	0.648
PEAK GPU MEMORY (GB)	4.37	5.94	4.41	3.25
SAMPLES PER SECOND	1.11	1.69	1.89	1.91
FID	111.12	111.59	112.75	110.73
PSNR (DB)	20.50	20.42	20.50	20.43

Bridging the Sim-to-Real Gap



Shaping the Future of Video Generation



World Model as An Environment for Policy Evaluation



Gripper Control



Sweep X-Axis



Sweep Y-Axis



Sweep Z-Axis

<https://world-model-eval.github.io/>

Our Scope

- Reimplement and optimize **Masked-HWM** for resource-constrained environments.
- Train the model on the original **1XGPT dataset**.
- Develop checkpoint loading, model export, and inference pipelines.
- Preprocess the **DROID** and **BridgeData V2** datasets for fine-tuning and evaluation.
- Benchmark pretrained and fine-tuned models through quantitative and qualitative analyses.

Why choose Masked-HWM?

The paper reports stronger practical performance for Masked-HWM than Flow-HWM

Masked-HWM example

Generated



Ground Truth



Flow-HWM example

Generated



Ground Truth



Reported quantitative comparison

Variant	FID ↓	PSNR ↑	Params	Samples/s
Masked-HWM Base	10.13	29.02	0.321B	2.27
Masked-HWM Sharing	14.21	28.66	0.195B	2.36
Flow-HWM Base	111.59	20.42	1.36B	1.69

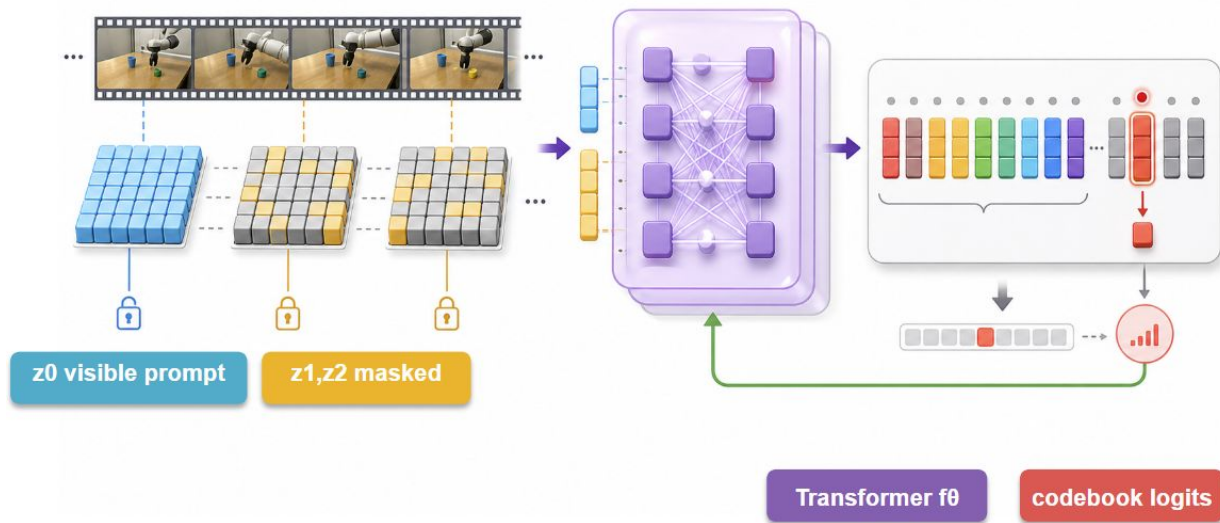
Masked-HWM delivers substantially better visual quality with fewer parameters and faster generation than Flow-HWM. Its discrete-token formulation is also more feasible to train and fine-tune under our limited Kaggle compute budget.

Implementation Details

Aspect	Paper / original setting	Our reimplementation
Tokenizer	Cosmos DV8x8x8 tokenizer	Cosmos DV8x8x8 tokenizer
Training target	Discrete VQ codebook IDs	Same: VQ codebook IDs
Temporal config	Original reports longer context/future setting	<code>T=3, num_prompt_frames=1</code>
Effective rollout	Predict future latent tokens	<code>z0</code> prompt → generate <code>z1, z2</code> → decode 17 RGB frames
Action conditioning	Video + action conditioning	<code>WITH_ACT=False</code> , visual-only
Loss	Cross-entropy on masked tokens	Same, CE on masked future tokens
Training steps	Reported 60k steps	10k-step Kaggle run / bounded runs
Hardware	A6000-scale training	Kaggle GPU session
Dataset	1X humanoid data	1X humanoid data + DROID & BridgeData V2

Training pipeline

Training happens in token space: the target is a Cosmos VQ codebook ID, not RGB pixels. (Image generated by codex session)



Loss on masked future tokens

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_{\theta}(y_i | x_{\text{masked}})$$

$\mathcal{M}$ = masked token positions in z_1, z_2

Gradient signal

$$\frac{\partial \mathcal{L}}{\partial t_i} = p_{\theta}(\cdot | x_{\text{masked}}) - \text{onehot}(y_i)$$

Target y_i is a VQ codebook ID, not a pixel.
CE pushes probability toward the correct token ID.
Visual-only batch: WITH_ACT=False.

Dataset Setup

Dataset	Role	Type	Used for	Novel?
1X-style	in-distribution	humanoid/egocentric	train + val	No
DROID	OOD-1	real robot manipulation	fine-tune + eval	Yes
BridgeData V2	OOD-2	real robot / simulated robot	fine-tune/eval	Yes

Evaluation Protocol

Evaluation of 2 checkpoints × 2 datasets (4 combinations in total):

- **Checkpoints:** 1X-style (base), Droid (fine-tuned on DROID)
- **Datasets:** val_1x_style (in-distribution), droid_ood (out-of-distribution)
- **Setup:** autoregressive rollout, 1 prompt frame → 2 predicted frames, MaskGIT steps = 8, temperature = 0, up to 200 samples per dataset

Token-space metrics (evaluate token prediction quality):

- CE Loss ↓
- Top-1 Token Accuracy ↑
- Perplexity ↓

Pixel-space metrics (evaluate the quality of generated images after decoding with Cosmos DV-8×8×8):

- PSNR ↑
- LPIPS ↓
- FID ↓

Efficiency:

- Generation time per frame (sec/frame)

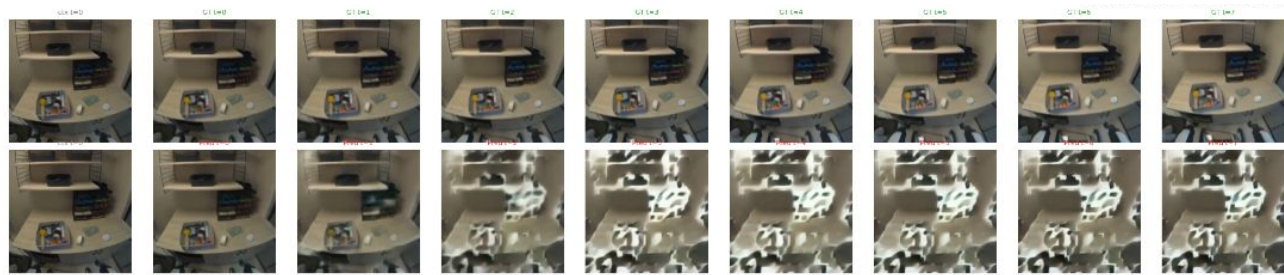
Quantitative Results

	Checkpoint	Dataset	# Samples	Loss	Accuracy	Perplexity	PSNR	FID	LPIPS	Generate time (sec/frame)
1	1X-style	val_1x_style	200	10.26	0.001	28440.32	14.06	286.07	0.42	1.56
2	1X-style	droid_ood	62	10.04	0.002	23018.06	13.54	269.0	0.46	1.56
3	1X-style	bridge_val	24	10.37	0.001	331782.21	13.36	331.59	0.55	1.546
4	Droid	val_1x_style	200	10.37	0.001	31846.97	15.10	284.50	0.39	1.577
5	Droid	droid_ood	62	9.88	0.002	19504.06	14.69	278.87	0.39	1.577
6	Droid	bridge_val	24	10.31	0.002	30085.39	13.94	304.15	0.51	1.572
7	Bridge	val_1x_style	200	10.35	0.001	31264.64	15.7	212.78	0.4	1.524
8	Bridge	droid_ood	62	10.18	0.001	26436.6	14.0	252.01	0.43	1.527
9	Bridge	bridge_val	24	9.9	0.004	19851.7	16.09	291.69	0.42	1.580

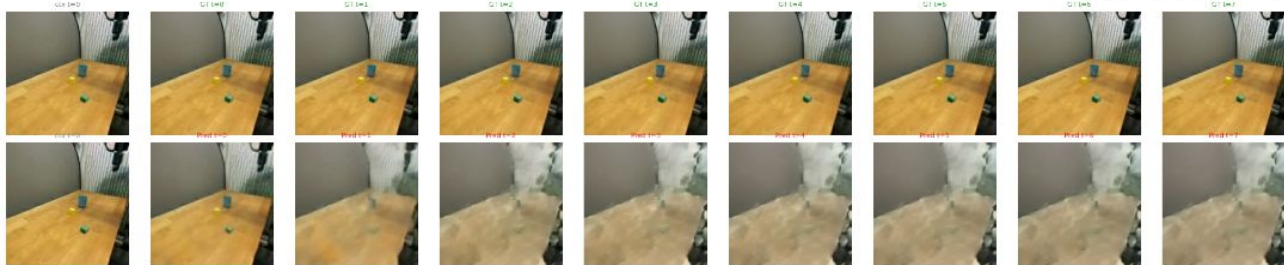
Qualitative Rollouts

1X-Style Checkpoint

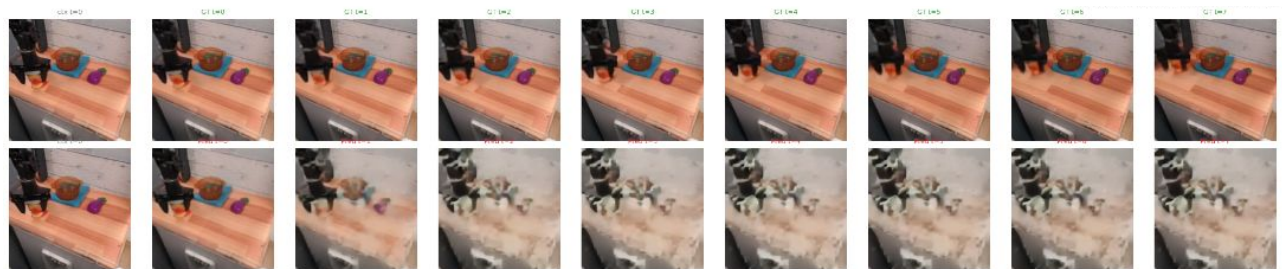
1X-style



DROID



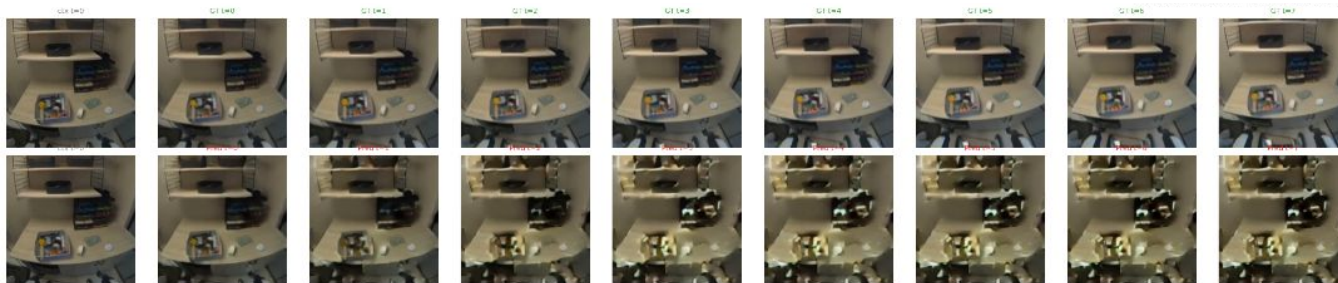
BRIDGE



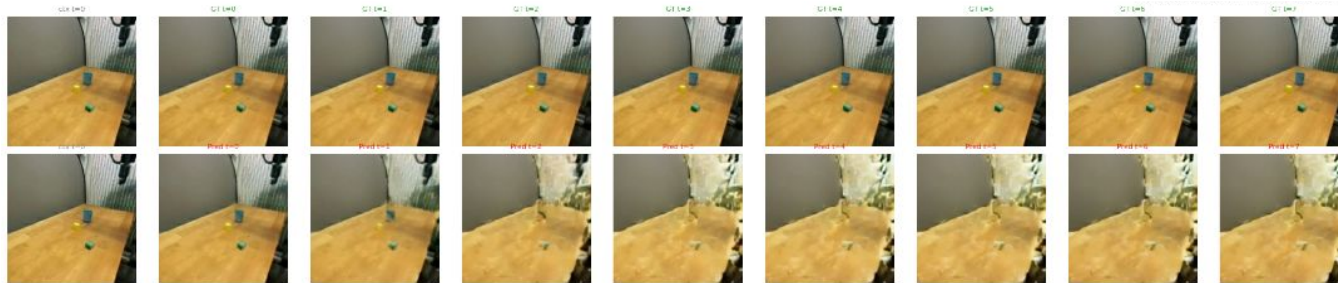
Qualitative Rollouts

DROID Checkpoint

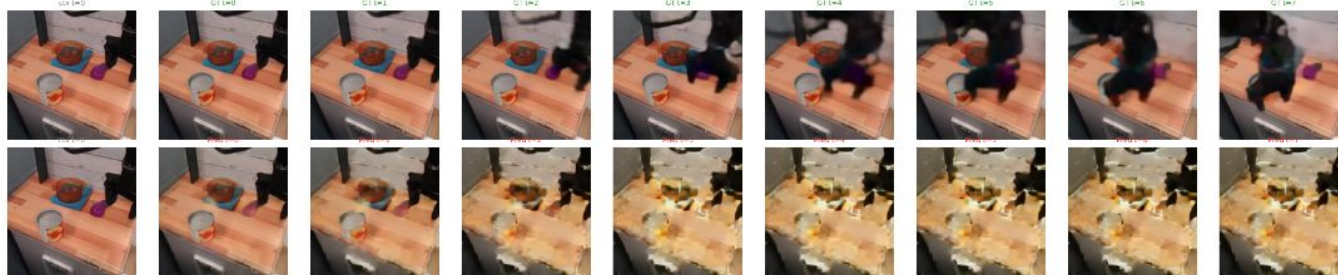
1X-style



DROID



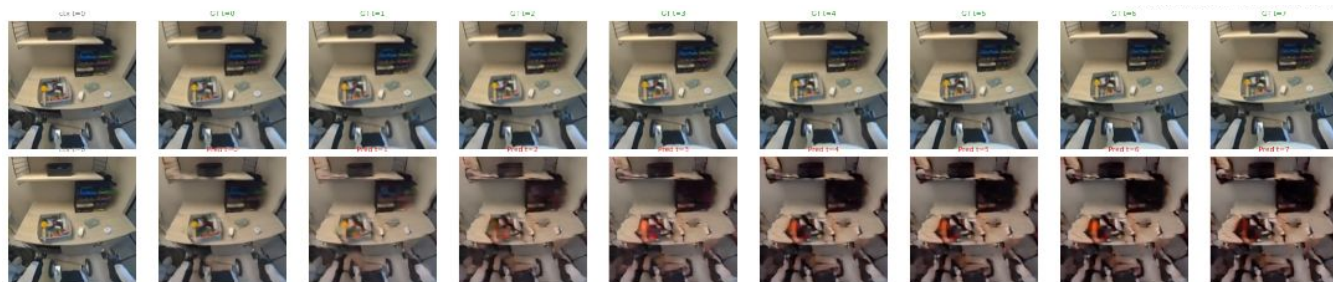
BRIDGE



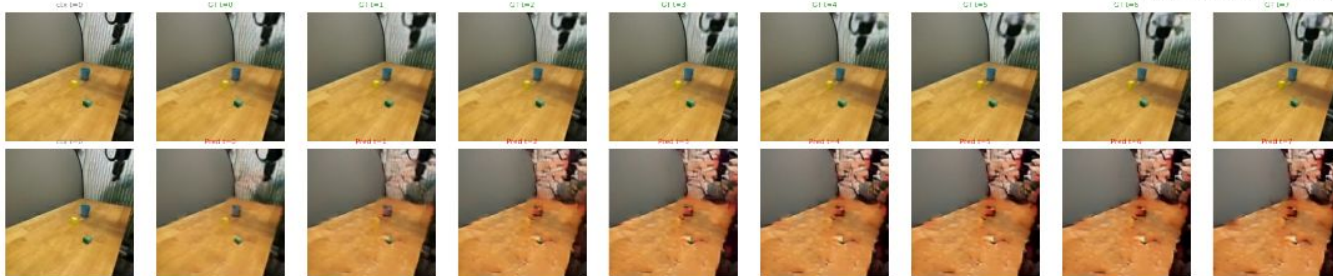
Qualitative Rollouts

BRIDGE Checkpoint

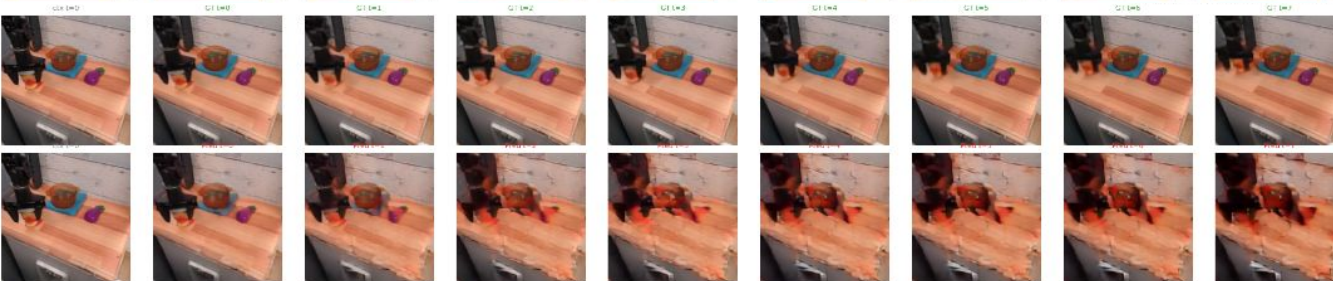
1X-style



DROID



BRIDGE



Qualitative Rollouts

Checkpoint 1X-style base

- On droid_ood: prediction collapses rapidly, only the first 2-3 frames retain recognizable structure (wooden table, green object), after which the entire sequence degrades into a uniform gray haze with no discernible scene content. The model has never been exposed to DROID-style scenes during training, leaving it with no meaningful prior to predict from.
- On val_1x_style: despite being in-distribution, predictions still degrade into bright white artifacts around frames 3-4. Object outlines fade progressively, and the scene loses structure earlier than expected.
- On bridge_val: degradation is even faster than the other two, recognizable structure (bowl, blue cutting board, purple object, wood table) survives only through t=0-1, and by t=2 the robot arm region has already smeared into a large white/gray blob that dominates the frame. From t=4-5 onward the entire scene has collapsed into the same uniform gray-white haze seen in droid_ood, with no object boundaries left. Despite bridge being closer to the training distribution than droid, the model doesn't hold structure any longer here than on the OOD set, likely because bridge scenes have more arm motion/occlusion near the camera, which the autoregressive rollout can't track and instead compounds error faster.

Qualitative Rollouts

Checkpoint DROID

- On droid_ood: clear improvement over the base model, the warm yellow/brown tones of the table surface and camera orientation are preserved considerably longer, with degradation into faint yellow artifacts only appearing toward the final frames. Overall scene structure remains recognizable through most of the rollout.
- On val_1x_style: retains scene outlines and darker tones longer than the base model, though dark artifacts begin emerging around the middle of the rollout. No signs of catastrophic forgetting, but no substantial improvement over the base model on this dataset either.
- On bridge_val: shows the same warm yellow/brown artifact pattern seen in droid_ood rather than the gray-white collapse of the base checkpoint, the wooden table tone and rough arm silhouette persist further into the rollout (through roughly $t=6-8$) before degrading into diffuse yellow-orange texture. However, fine object details (bowl, blue mat, purple item) are lost earlier than the table color/tone is, and the arm shape becomes a smeared yellow blob rather than a gray one. This suggests the DROID finetuning shifted the model's "failure color" toward DROID's own domain statistics (warm wood tones) rather than genuinely improving structural persistence, the collapse still happens, it's just repainted in DROID's palette, which shows up clearly on bridge_val since bridge's table is a similar warm wood tone to droid's.

Qualitative Rollouts

Checkpoint BRIDGE

- On droid_ood: similar to val_world_model, the warm wood tone of the droid table is preserved reasonably long (through roughly $t=8-10$), but the same red/orange blotches overlay the surface, concentrated especially around the robot arm and table corner. Compared to the DROID checkpoint (uniform yellow/brown convergence), this artifact is noisier and higher-contrast, resembling amplified noise texture accumulating through the autoregressive steps rather than a single convergent color.
- On val_1x_style: holds scene layout (desk, tool tray, small items) considerably better than the previous two checkpoints, object positions and shapes remain recognizable well into the rollout. However, a new distinctive artifact emerges: red/orange/black streaks bleed across the table surface starting around $t=3-4$, spreading into "burnt-looking" patches that dominate large portions of the frame by the later steps, even though the overall desk outline and background structure are still intact.
- On bridge_val: the same red/orange artifact pattern appears earliest and most prominently here of the three datasets, as early as $t=2-3$, red streaks already surround the bowl and robot-arm region, and by $t=6$ onward nearly the entire frame is covered in chaotic orange/red noise texture, though the wooden table and bowl silhouette are still faintly traceable at the edges.

Comparison with Paper - All

Aspect	Paper	Our reproduction
Model family	Masked-HWM + Flow-HWM	Masked-HWM only
Conditioning	video + actions	video-only if <code>WITH_ACT=False</code>
Dataset	100h humanoid demos	original subset + DROID + Bridge
Metrics	FID, PSNR, speed, memory	CE, acc, PPL, PSNR, LPIPS, FID, time
Main finding	Masked-HWM > Flow-HWM under constraints	test if Masked-HWM transfers to novel robot datasets

Final Takeaways & Limitations

Takeaways

- Reimplemented the Masked-HWM token-prediction pipeline on Kaggle.
- Trained and ran inference on the 1XGPT tokenized dataset.
- Built a preprocessing path from raw robot videos to HWM-compatible tokens.
- Verified DROID compatibility with the visual-token loader.
- Prepared zero-shot / fine-tune evaluation under domain shift.

Limitations

- No full Flow-HWM reproduction.
- Visual-only setting: action conditioning is disabled.
- DROID / Bridge are robot manipulation datasets, not humanoid egocentric data.
- Training runs are bounded by Kaggle compute, not paper-scale hardware.
- Second novel dataset and LPIPS/FID should be reported only if fully executed.